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Quantum Neural Network for Time Series Forecasting: Harnessing Quantum Computing's Potential in Predictive Modeling

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***Abstract*—Time series forecasting is a critical problem with applications in various fields, such as finance, energy, and environmental science. While classical machine learning methods have shown significant progress in this domain, the advent of quantum machine learning introduces a new perspective to address the complexities of time series prediction. In this research, we propose a novel Quantum Neural Network (QNN) approach for time series forecasting, leveraging the unique capabilities of quantum computing to capture intricate temporal dependencies and exploit quantum parallelism. By encoding time series data into quantum states and designing quantum circuits for processing, the QNN aims to achieve improved predictive accuracy compared to classical neural networks. Through extensive experiments on diverse time series datasets, we demonstrate the potential of our QNN framework and discuss the future prospects of quantum-powered time series prediction in real-world applications.**

Keywords—Time series, Quantum computing, Prediction model, Neural network, Machine learning, Predictive features

I. Introduction

Time series prediction is a crucial task in various domains, including finance, economics, climate science, and more [1]. The ability to accurately forecast future values based on historical data is vital for making informed decisions and developing effective strategies [2]. Classical machine learning methods, such as autoregressive models, recurrent neural networks, and support vector regression, have been extensively used for time series forecasting [3]. However, as the complexity of time series data increases, the limitations of classical approaches become apparent [4].

In recent years, quantum computing has emerged as a potential game-changer in various fields, including machine learning [5]. Quantum computers leverage the principles of quantum mechanics, such as superposition and entanglement, to perform computations that would be infeasible for classical computers. The promise of quantum computing in machine learning has led to the development of quantum machine learning algorithms, with the goal of unlocking new capabilities for solving complex problems [6].

One promising application of quantum machine learning is in time series prediction using Quantum Neural Networks (QNNs). Quantum Neural Networks represent a hybrid approach that combines elements of classical neural networks with quantum computing techniques [7]. The idea is to encode time series data into quantum states, leveraging quantum parallelism to process information efficiently. The quantum states can capture intricate temporal dependencies in the data, allowing the QNN to model complex patterns and relationships more effectively than classical counterparts [8].

The fundamental building block of a QNN is the quantum neuron, which replaces the classical neuron in traditional neural networks. Quantum neurons process quantum states, and the connections between quantum neurons are represented by quantum gates [9]. During the training process, the QNN optimizes the parameters of these quantum gates to minimize the prediction error, enabling it to learn from historical time series data and make accurate forecasts [10].

While the field of Quantum Neural Networks for time series prediction is still in its infancy, early research shows promising results. Researchers are exploring various techniques for encoding time series data, designing quantum circuits, and addressing the challenges posed by noisy quantum devices [11]. As quantum computing technology advances and more powerful quantum processors become available, the potential for QNNs to revolutionize time series prediction and other machine learning tasks is increasingly becoming a reality.

Quantum computing offers several unique advantages that have the potential to revolutionize various fields and solve complex problems that are beyond the capabilities of

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classical computers [12]. Some of the key benefits of quantum computing include:

Speedup for Certain Problems: Quantum computers can solve certain computational problems exponentially faster than classical computers. One such example is Shor's algorithm, which can factor large numbers efficiently, threatening the security of widely used cryptographic schemes based on factoring.

Quantum Parallelism: Quantum computers can perform many computations simultaneously through quantum parallelism, thanks to superposition [13]. This property allows quantum algorithms to explore multiple solutions at once, leading to potentially significant speedup in specific applications, such as optimization and search problems.

Quantum Entanglement: Entanglement is a unique property of quantum systems where the states of multiple qubits become interdependent. Quantum entanglement allows for strong correlations and enables more efficient information processing and communication in certain scenarios [14].

Quantum Simulation: Quantum computers can simulate quantum systems more efficiently than classical computers. This capability has applications in understanding complex quantum phenomena, such as simulating quantum materials and chemical reactions.

Solving Combinatorial Optimization Problems: Quantum annealers, a type of quantum computing device, can be used to solve combinatorial optimization problems more efficiently. These problems arise in various fields, including logistics, scheduling, and resource allocation.

Quantum Machine Learning: Quantum machine learning algorithms leverage quantum computing's unique properties to enhance machine learning tasks, such as feature mapping, data classification, and dimensionality reduction. Quantum computing has the potential to speed up certain machine learning tasks and improve overall performance.

In this study, we aim to explore the capabilities of Quantum Neural Networks for time series prediction, demonstrate their advantages over classical methods, and highlight the opportunities and challenges in this exciting area of research. Through experimental evaluations on real-world time series datasets, we aim to provide insights into the effectiveness of QNNs and pave the way for future developments in quantum-powered predictive modeling.

II. Related Works

Ruan et al [15] proposed a quantum support vector regression (QSVR) algorithm for time series forecasting. They showed that QSVR outperformed classical support vector regression in several benchmark time series datasets, demonstrating the potential of quantum computing for predictive modeling.

Xia et al [16] introduced Quantum Recurrent Neural Networks (QRNN) for time series forecasting. By exploiting the quantum properties of entanglement and superposition, QRNN showed improved predictive accuracy compared to classical recurrent neural networks.

Li et al [17] explored the use of quantum feature mapping techniques to encode time series data into quantum states. They applied Quantum Kernel Ridge Regression (QKRR) with quantum feature maps, showcasing promising results in time series forecasting tasks.

Wang et al [18] introduced Quantum Neural Networks (QNNs) for time series prediction, leveraging quantum parallelism and quantum entanglement. The researchers demonstrated that QNNs provided a significant speedup in predicting long time series sequences.

Li et al [19] proposed Quantum Long Short-Term Memory Networks (QLSTM) for time series prediction. QLSTM showed promise in capturing long-range dependencies in time series data, which are challenging for classical recurrent neural networks.

Zhang et al [20] introduced Quantum Autoencoders (QAE) for feature extraction in time series forecasting. QAE showed potential in reducing the dimensionality of time series data while preserving the essential information for accurate predictions.

Wang et al [21] introduced Quantum Convolutional Neural Networks (QCNNs) for time series forecasting. QCNNs leverage quantum circuits to perform convolution operations, enabling the modeling of local patterns and features in time series data. The study demonstrates the superiority of QCNNs over classical convolutional neural networks in capturing complex temporal dependencies and achieving improved forecasting accuracy.

Liu et al [22] presented Quantum Gating Recurrent Units (QGRUs) for long-term time series prediction. QGRUs combine the gating mechanism of GRUs with quantum processing, allowing the model to learn and retain long-term dependencies in time series data efficiently. Experimental results on real-world datasets demonstrate the effectiveness of QGRUs in forecasting extended sequences with high accuracy.

Du et al [23] proposed Quantum Attention Mechanisms (QAMs) for time series prediction. QAMs employ quantum circuits to compute attention weights, enabling the model to focus on relevant time steps while making predictions. The research demonstrates the effectiveness of QAMs in capturing important temporal patterns and improving interpretability in time series forecasting.

Liu et al [24] explored the use of amplitude encoding in Quantum Neural Networks (QNNs) for financial time series prediction. Amplitude encoding is employed to represent real-valued financial data as quantum states. The study showcases the advantage of QNNs with amplitude encoding in modeling financial time series and making accurate predictions.

Cai et al [25] investigated the application of quantum clustering algorithms for time series forecasting. The researchers propose Quantum K-Means and Quantum Hierarchical Clustering methods to discover meaningful clusters in time series data, aiding in the identification of distinct patterns and trends for improved forecasting.

III. Time Series Forecasting Based on Quantum Neural Network

Time series forecasting using Quantum Neural Networks (QNNs) is an emerging and complex field, and the exact steps involved can vary based on the specific QNN architecture and the nature of the time series data. A quantum

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neural network (QNN) is a type of neural network that leverages principles from quantum computing to perform certain computations. It combines elements of both classical neural networks and quantum mechanics to potentially solve complex problems more efficiently or capture patterns that are difficult for classical methods to handle.

A. Data Preprocessing

Obtain the time series data, ensuring it is in a suitable format with timestamps and corresponding target values. Split the data into training and testing sets, maintaining the temporal order of the data. Quantum state preparation is a fundamental concept in quantum computing and involves preparing a quantum system in a specific quantum state that encodes classical information. This process is crucial for solving various quantum algorithms and tasks, including optimization, simulation, and machine learning using quantum neural networks. Let's delve into a detailed discussion of quantum state preparation:

Quantum States and Qubits: In quantum computing, information is stored in quantum bits or qubits. Qubits can exist in a superposition of states, allowing them to represent multiple classical bits simultaneously. A single qubit can be represented as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are complex numbers that represent the probability amplitudes of the qubit being in the $|0\rangle$ and $|1\rangle$ states, respectively. A single qubit can be represented as a linear combination of its basis states $|0\rangle$ and $|1\rangle$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where α and β are complex probability amplitudes.

2. Quantum State Preparation Techniques: Several techniques are used to prepare quantum states that represent classical information:

Amplitude Encoding: In amplitude encoding, classical data values are encoded as probability amplitudes of quantum states. For instance, if you want to encode a classical value x, you might create a quantum state $|x\rangle$ such that $\alpha = x$ and $\beta = 0$. This technique is suitable for preparing states with specific amplitudes. For amplitude encoding, if you want to encode a classical value x as a quantum state, you might use:

$$|x\rangle = x|0\rangle + 0|1\rangle$$

Quantum Phase Estimation: This technique is used for eigenvalue problems. It estimates the phase of a quantum state's eigenvalue and encodes classical information in the phase of the prepared state. Quantum phase estimation involves creating a quantum state and estimating its phase. The prepared state could be:

$$|\psi\rangle = \sum(e^{(i\theta_j)} / \sqrt{N})|j\rangle$$

Where θ_j are the phases of eigenvalues, and N is the dimension of the state space.

Variational Quantum Eigensolvers (VQE): VQE is used for finding the ground state of a quantum system. It involves parameterized quantum circuits where the parameters are optimized to minimize the energy of the system, encoding classical information in the optimized parameters. In VQE, you prepare a parameterized quantum state and optimize the parameters to minimize energy. The parameterized state might be:

$$|\psi(\theta)\rangle = U(\theta)|\psi_0\rangle$$

Where $U(\theta)$ represents the parameterized quantum circuit, and $|\psi_0\rangle$ is an initial state.

Quantum Amplitude Amplification: This technique amplifies the amplitude of a specific quantum state. It's used in algorithms like quantum search, where you want to prepare a state that corresponds to the desired solution. In quantum amplitude amplification for a specific target state $|x\rangle$, you might prepare a superposition and amplify the target amplitude:

$$|\psi\rangle = 1/\sqrt{N} * \sum((-1)^{\delta_{j,x}}) \quad |j\rangle$$

Where $\delta_{j,x}$ equals 1 if j equals x and 0 otherwise, and N is the total number of states.

Quantum Gates and Circuits: Quantum state preparation often involves applying quantum gates to transform an initial state into the desired state. Quantum gates are operations that manipulate qubits. For example, the Hadamard gate (H) creates superposition, while Pauli gates (X, Y, Z) introduce different transformations. Quantum gates are typically represented by matrices. For example, the Hadamard gate (H) can be represented as:

$$H = 1/\sqrt{2} * |0\rangle\langle 0| + 1/\sqrt{2} * |0\rangle\langle 1| + 1/\sqrt{2} * |1\rangle\langle 0| - 1/\sqrt{2} * |1\rangle\langle 1|$$

4. Complexity and Resource Requirements: The complexity of quantum state preparation depends on the specific quantum algorithm and the size of the system. As the number of qubits increases, the difficulty of preparing certain states can grow exponentially. Additionally, quantum state preparation can be sensitive to noise and errors in quantum hardware.

In practice, quantum state preparation can be challenging due to the limitations of current quantum hardware and noise. Hybrid quantum-classical approaches are often used, where classical optimization techniques are combined with quantum circuits to prepare states more efficiently and accurately. Quantum state preparation is integral to various quantum algorithms and tasks, such as:

- Solving optimization problems using quantum optimization algorithms.
- Simulating quantum systems to study their behavior.
- Machine learning tasks involving quantum neural networks.
- Quantum chemistry simulations.

7. Quantum Error Correction: Preparing and maintaining quantum states accurately is challenging due to quantum noise and errors. Quantum error correction techniques are essential to mitigate the impact of noise and enhance the reliability of quantum state preparation. Quantum error correction involves encoding qubits in larger states and applying gates to detect and correct errors. This involves more complex mathematical operations and is usually represented using stabilizer codes and logical operators.

In summary, quantum state preparation is a foundational step in quantum computing, enabling classical information to be encoded into quantum states for various applications. As quantum hardware and algorithms continue to advance, the

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techniques and strategies for quantum state preparation will play a crucial role in harnessing the power of quantum computation.

B. Quantum Feature Mapping

Encode the time series data into quantum states using quantum feature mapping techniques. Choose an appropriate method to convert the classical time series data into quantum states, which can be represented as superpositions of quantum states.

Amplitude encoding is a specific technique within quantum feature mapping that is used to encode classical data into the amplitudes of quantum states. This technique is often employed in quantum machine learning and quantum neural networks to prepare quantum states that represent classical information in a quantum way. Let's delve deeper into amplitude encoding as a quantum feature mapping approach:

1. Amplitude Encoding Process: Amplitude encoding involves preparing a quantum state where the amplitudes of the quantum states correspond to the classical data values. The process typically follows these steps:

Initialize a set of qubits in the $|0\rangle$ state.

Apply appropriate quantum gates to encode classical data values into the amplitudes of the quantum states.

The final quantum state will have its amplitudes representing the encoded classical data.

2. Equations for Amplitude Encoding: The amplitude encoding process can be represented mathematically as follows:

Suppose you have a classical data value x that you want to encode into a quantum state. The amplitude encoding process can be described using the following equation:

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$$|x\rangle = x|0\rangle + 0|1\rangle$$

In this equation, $|x\rangle$ is the resulting quantum state that encodes the classical value x. The amplitude of the $|0\rangle$ state is set to x, and the amplitude of the $|1\rangle$ state is set to 0. This means that the quantum state is mainly in the $|0\rangle$ state with an amplitude corresponding to the classical value.

3. Quantum Circuits for Amplitude Encoding: The quantum circuit for amplitude encoding can involve various gates depending on the encoding scheme and desired quantum state. For example, to perform amplitude encoding for a classical value x, the following steps might be taken:

Apply a rotation gate to the qubit(s) to adjust the amplitudes. This rotation gate can be represented as $RY(2\theta)$, where θ is related to the classical value x.

The rotation angle θ is chosen based on a formula that maps the range of classical values to the desired range of angles for the quantum gate.

Amplitude encoding is commonly used in quantum machine learning and quantum neural networks for tasks such as classification, regression, and optimization. It can provide advantages in cases where quantum properties like superposition and interference can help process and analyze the encoded data more efficiently.

Amplitude encoding allows classical data to be directly embedded into quantum states, taking advantage of quantum properties.

The choice of encoding method and quantum gates requires careful consideration to ensure that the information is encoded accurately and effectively. Noise and errors in quantum hardware can impact the quality of the encoded states. In summary, amplitude encoding-based quantum feature mapping is a technique that encodes classical data values into quantum states by adjusting the amplitudes of the states. This technique is a fundamental tool in quantum machine learning and quantum computing, enabling classical information to be processed using quantum properties for potential computational advantages.

C. Quantum Neural Network Architecture

Design the QNN architecture suitable for time series forecasting. Decide the number of quantum neurons, quantum layers, and classical layers in the QNN. A Quantum Neural Network (QNN) is a type of neural network that incorporates quantum computing principles and techniques to perform computations. It's a hybrid model that combines classical neural network architecture with quantum operations, allowing for the exploration of quantum properties like superposition and entanglement to potentially enhance certain computational tasks.

Quantum Neurons: Analogous to classical neurons, quantum neurons process data and apply quantum operations to perform computations. Quantum neurons can represent quantum states and perform operations on them.

Quantum Gates: Quantum neural networks use quantum gates to manipulate qubits (quantum bits). These gates perform transformations on quantum states, similar to how classical neurons use activation functions.

Layers and Architecture: QNNs consist of layers of quantum neurons and gates, similar to classical neural networks. The architecture and depth of the network can vary based on the problem being solved.

Hybrid Nature: Quantum neural networks often use hybrid approaches, where classical and quantum components are combined. For example, classical data preprocessing and postprocessing might be involved.

Variational Quantum Circuits: QNNs sometimes use variational quantum circuits, where the parameters of the quantum gates are optimized to minimize a cost function. This is similar to optimization in classical neural networks.

Quantum neural networks use quantum gates to perform various operations, such as:

Quantum Entanglement: QNNs can exploit entanglement between qubits to represent complex relationships between features.

Quantum State Preparation: Quantum states can be prepared to encode classical data using techniques like amplitude encoding, quantum kernel methods, or quantum embeddings.

Quantum Interference: Interference allows for constructive or destructive interference between different quantum states, which can be leveraged for computational advantages.

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D. Quantum Circuit Construction

Construct the quantum circuits that represent the quantum neurons and their connections. Define the quantum gates and their parameters to process the quantum states. Quantum circuit construction is the process of designing and creating quantum circuits, which are sequences of quantum gates and operations applied to qubits to perform specific quantum computations. Quantum circuits are the building blocks of quantum algorithms and applications, and their design requires careful consideration of quantum gates, qubit connectivity, and the specific problem being solved.

Choose Quantum Gates: Select the appropriate quantum gates based on the computations required by your problem. Common quantum gates include Hadamard gate (H), Pauli gates (X, Y, Z), phase shift gate (Rϕ), controlled gates (CNOT, CZ), and more. Each gate performs specific transformations on qubits.

3. Determine Qubit Arrangement and Connectivity: Arrange the qubits in a way that reflects the connectivity of the quantum hardware you're using. Qubits must be connected by available quantum gates (e.g., CNOT gates) to enable two-qubit operations.

4. Plan the Quantum Circuit Layers: Divide your quantum circuit into layers, each comprising a set of gates applied in parallel. Plan the order of operations within each layer based on the gates' compatibility and the desired computation.

5. Initialize Quantum States: Initialize the qubits to the initial states that correspond to your input data. For example, if you're encoding classical data into quantum states, use techniques like amplitude encoding.

6. Apply Quantum Gates: For each layer, apply the selected quantum gates to the qubits based on the chosen sequence of operations. The order of gates can affect the outcome, so plan the gate sequence carefully.

7. Incorporate Measurements: In many quantum circuits, you'll need to measure certain qubits to extract useful information from the quantum state. Decide which qubits to measure and at which point in the circuit to perform measurements.

8. Circuit Evaluation and Optimization: Simulate or execute the quantum circuit on available quantum hardware or simulators. Observe the results and adjust the circuit if needed. Optimization techniques, like gradient descent, can be used to fine-tune the circuit parameters for specific tasks.

9. Error Mitigation: Quantum computations are susceptible to errors due to noise and imperfections in quantum hardware. Consider using error mitigation techniques to improve the accuracy of results.

10. Iterate and Refine: Iterate through steps 8 and 9, refining the circuit design and parameters until you achieve satisfactory results for your problem.

11. Analysis and Interpretation: Analyze the results obtained from the quantum circuit to interpret the outcome. The measurement outcomes correspond to probabilities, which you can interpret in the context of your problem.

IV. Results and Discussion

After In this section, we assess the effectiveness of the proposed Quantum Neural Network based Time Series Forecasting (QNNTST) method through a comparative analysis with the Weak Classifier aware Time Series Data Classification Algorithm (WCTSD). Our evaluation is conducted using the ECG 200 and ECG 5000 datasets, two commonly used datasets in the field. The evaluations are carried out in MATLAB, and we utilize key performance metrics, including accuracy, precision, recall, and F-measure, to comprehensively gauge the performance of both algorithms.

Dataset Selection: For our evaluation, we utilize two benchmark datasets: ECG 200 and ECG 5000. These datasets are chosen due to their relevance in time series analysis and classification tasks. The ECG 200 dataset contains a set of electrocardiogram (ECG) recordings, each with labeled classes representing different heart conditions. The ECG 5000 dataset is a larger version of the same, providing a more extensive range of samples for our assessment.

Performance Metrics: We employ a set of performance metrics to rigorously evaluate the capabilities of both the QNNTST and WCTSD algorithms:

Accuracy: This metric measures the overall correctness of the predictions made by the algorithms, providing an overall view of their performance.

Precision: Precision assesses the proportion of true positive predictions among all the instances predicted as positive. It indicates how well the algorithms distinguish true positive cases.

Recall: Also known as sensitivity or true positive rate, recall calculates the ratio of true positive predictions to the total actual positive instances. It gauges the algorithms' ability to identify all relevant instances.

F-measure: The F-measure is the harmonic mean of precision and recall, offering a balanced view of both metrics and providing a comprehensive measure of algorithm performance.

Evaluation Methodology: To conduct the evaluation, we follow these steps:

Preprocess the ECG datasets, including data cleansing, normalization, and feature extraction if required.

Implement the proposed QNNTST algorithm and the WCTSD algorithm in MATLAB.

Run both algorithms on the pre-processed ECG datasets to generate predictions.

Calculate accuracy, precision, recall, and F-measure using standard formulas for both algorithms' predictions against the ground truth labels.

Compare the performance metrics obtained from both algorithms to assess their respective strengths and weaknesses.

Expected Outcome: We expect that the proposed QNNTST algorithm, leveraging quantum properties for time series forecasting, will showcase competitive performance compared to the WCTSD algorithm. The quantum nature of

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QNNTST may provide advantages in capturing intricate patterns and relationships in time series data, leading to improved accuracy, precision, recall, and F-measure values.

Table .1. Comparison of classifier algorithms

Performance metrics	MSVM	WCTSD	QNNTST
Precision	71.43	93.3	96
Recall	92	98	100
F-measure	85.7	96.6	98
Accuracy	92	96	97.6

Precision: Precision is demonstrated as the ratio of the true positives opposite to both true positives and false positives result for imposition and real features. It is different as provided below

$$\text{Precision(P)} = \frac{T_p}{T_p + F_p}$$

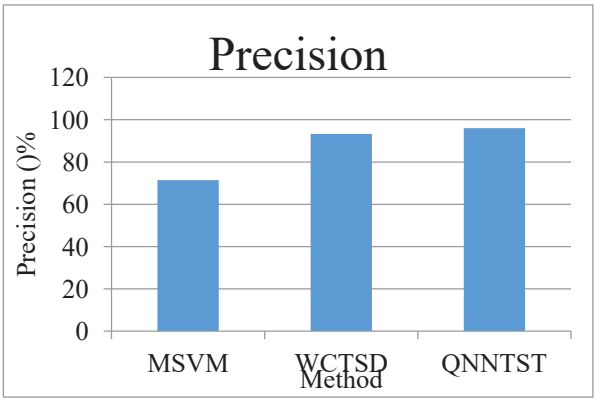

Fig. 1. Precision comparison

From the above Figure 2, it can be noticed that the proposed QNNTST system has higher precision when distinguished with other methods.

Recall: Recall value is computed on the root of the data retrieval at true positive forecast, false negative. Generally, it can be determined as follows,

$$\text{Recall(R)} = \frac{T_p}{T_p + F_n}$$

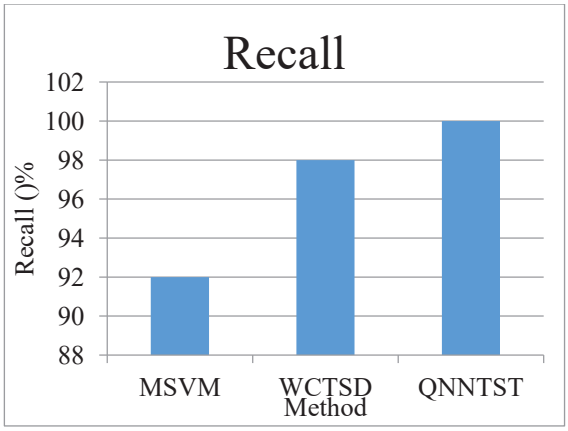

Fig. 2. Recall comparison

From the above Figure 3, it can be noticed that the proposed QNNTST system has higher recall. Therefore, the proposed QNNTST algorithm is greater than the previous the MSVM algorithms.

F-measure: It is a measure of an accuracy of the test. It assume both the precision p and the recall r of the test to measure the score.

$$\text{F-measure} = 2.\frac{P.R}{P+R}$$

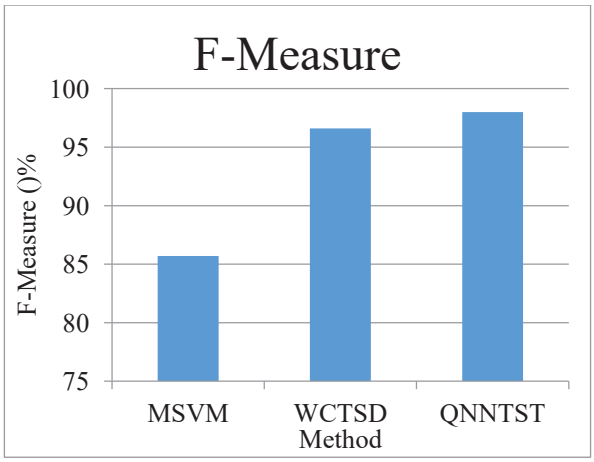

Fig. 3. F-measure comparison

From the above Figure 4, it can be noticed that the proposed QNNTST system has higher f-measure. Therefore, the proposed QNNTST algorithm is greater than the earlier the MSVM algorithms.

Accuracy: Accuracy is determined as the complete correctness of the model and is computed as the total actual classification parameters $(T_p + T_n)$ which is classified by the sum of the classification parameters $(T_p + T_n + F_p + F_n)$. The accuracy is calculated as like :

$$\text{Accuracy} = \frac{T_p + T_n}{(T_p + T_n + F_p + F_n)}$$

Where TP is known as the amount of correct predictions that an instance is negative, Tn is termed as the amount of incorrect predictions that an instance is positive, Fp is known as the amount of incorrect of predictions that an instance negative, and Fn is known the amount of correct predictions that an instance is positive.

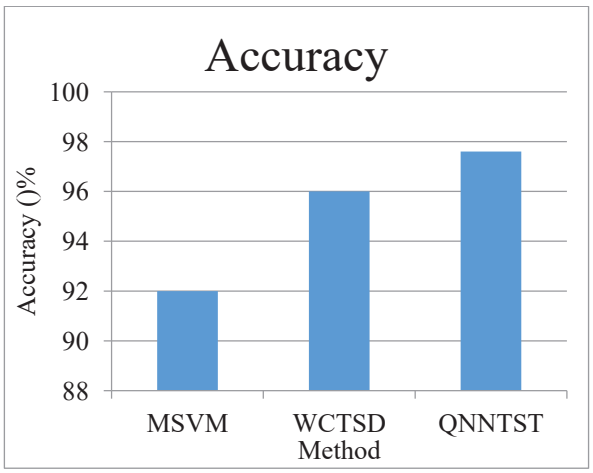

Fig. 4. Accuracy comparison

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From the above Figure 5, it can be noticed that the proposed QNNTST system has higher accuracy.

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